PROGRAM 10: AI-Based Research Assistant

AIM

To develop an AI research assistant that generates information, keywords and a concise summary for a given research topic.

ALGORITHM

1. Import the required library.
2. Load a pretrained FLAN-T5 language model.
3. Accept a research topic.
4. Generate information about the topic.
5. Extract relevant keywords.
6. Generate a concise summary.
7. Display all results.

CODE

```
from transformers import pipeline


research_model = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)


topic = input("Enter research topic: ")


# Generate information
info_prompt = (
    "Explain the research topic in simple words: " + topic
)


information = research_model(
    info_prompt,
    max_new_tokens=120
)[0]["generated_text"]
```

```
# Generate keywords
keyword_prompt = (
    "Give five important keywords related to this topic: " + topic
)

keywords = research_model(
    keyword_prompt,
    max_new_tokens=50
)[0]["generated_text"]

# Generate summary
summary_prompt = (
    "Write a short summary about this topic: " + topic
)

summary = research_model(
    summary_prompt,
    max_new_tokens=80
)[0]["generated_text"]

print("\nResearch Information:")
print(information)

print("\nResearch Keywords:")
print(keywords)

print("\nResearch Summary:")
print(summary)
```

INPUT

Enter research topic:

Applications of Artificial Intelligence in Healthcare

OUTPUT

Research Information:

Information about applications of AI in healthcare.

Research Keywords:

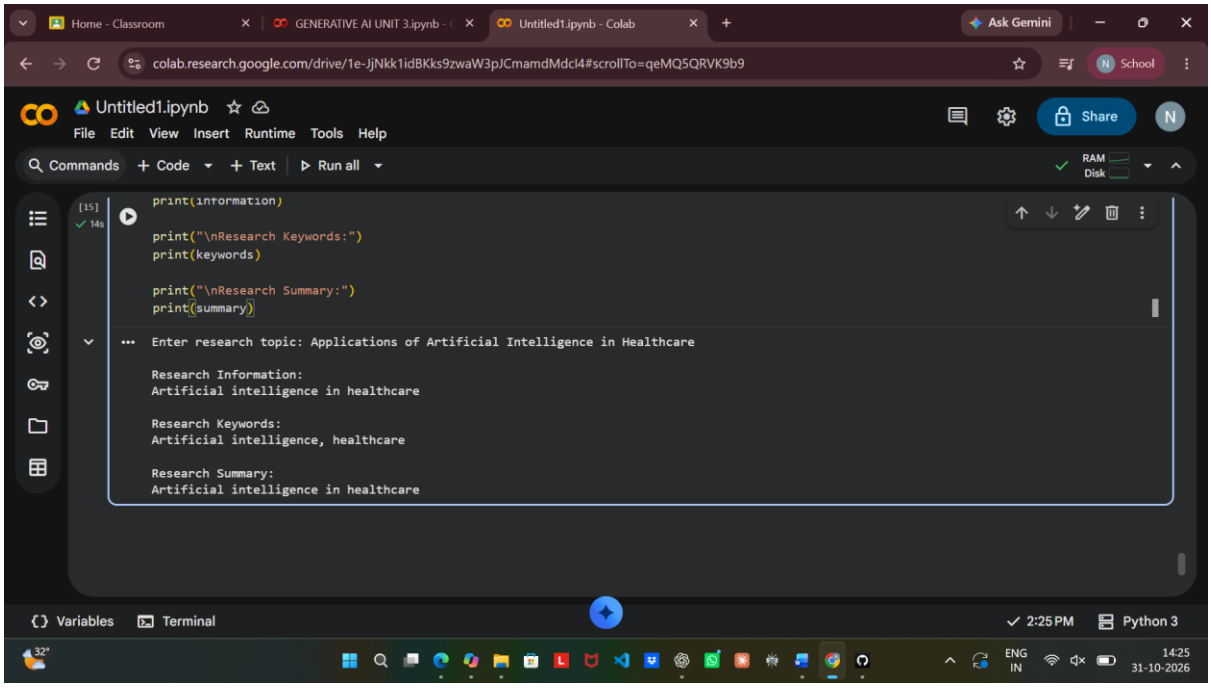
Artificial Intelligence, Healthcare, Machine Learning,

Medical Diagnosis, Deep Learning.

Research Summary:

A short summary about the applications of AI in healthcare.

The actual generated information will vary.



RESULT

Thus, an AI-based research assistant was implemented successfully using a pretrained language model.